Part B · Statistics

This is the second half of your folio. Your teacher marks it together with the Part A you did last term.

Name		Class		Date	

Our team	

The data I am using	our team / another team / spare data

You work on your own. You may use your team's cleaned file, your decision log, your displays from earlier lessons, the newsletter article, a calculator and the conversion mat. You may not use the Lesson 14 worksheet, the protocol cards, or your teammates. If your file will not open, put your hand up straight away.

There are seven items. B1 and B3 you have done before with this data. B6 is the longest piece of writing in the whole folio — leave yourself about fifteen minutes for it.

Our investigation

SF B1

State your statistical question. Then name every variable your team collected and say what type each one is.

OUR STATISTICAL QUESTION

A statistical question names a group, a thing you measured, and how you measured it. Starter: How much did the _____ who _____ enjoy _____?

Variable	Its type	How I know it is that type

The four types: nominal (groups, no order) · ordinal (groups, in an order) · discrete (counted) · continuous (measured). The test for the last two: does half of one make sense?

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How we collected it

CF B2

Describe how your data was collected. Then identify one limitation of the way you did it, and say what difference that limitation could make to your results.

HOW WE COLLECTED THE DATA

ONE LIMITATION, AND WHAT DIFFERENCE IT COULD MAKE

Your decision log will help. Starters: One thing that went wrong was _____. This could mean our results _____.

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One categorical variable

SF B3

Build a display for one categorical variable. Sketch it below, then describe it by its mode and its shape.

The variable I chose	

Give it a title and label both axes.

THE MODE

THE SHAPE — A SENTENCE ABOUT THE BUDDIES, NOT ABOUT THE BARS

Starters: The most common _____ was _____. · Most buddies _____, and _____.

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Grouping the measurements

CF B4

Group your continuous data into intervals, build a display, and describe the shape of the grouped data.

Group (seconds)	How many
26 to 29	
30 to 33	
34 to 37	
38 to 41	
42 to 45	
Total	

These groups are given to you. Check your total matches the number of measurements you actually have.

Bars touching, because the groups run into each other. Title it and label both axes.

THE SHAPE OF THE GROUPED DATA

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Comparing two distributions

CF B5

Compare two distributions. Name the feature that differs and say what it tells you.

The two things I am comparing	

	The first one	The second one
How many altogether		
Mode		
The feature I am comparing		

MY COMPARISON — FEATURE, BOTH VALUES, WHAT IT MEANS

Starter: _____ was higher than _____, because _____ out of _____ compared with _____ out of _____. This means _____.

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The newsletter article

CU B6

Critique the article. Identify what is misleading about it and explain the effect each fault would have on somebody reading it.

Every sentence in the article is true. Do not look for a lie — look for what it makes a reader believe, and whether the numbers support that. Naming a fault is the start of the answer. Saying what it makes somebody believe is the answer.

What is misleading	What it makes a reader believe	What the numbers actually show

SHOULD THIS ARTICLE BE PUBLISHED? WHY?

Starters: This makes a reader think _____. What is actually true is _____. I think the article should / should not go out because _____.

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Redrawing one graph

CF B7

Redraw one graph from the article fairly. Then explain what changed, and what did not change.

Same numbers. Title it and label the axis.

WHAT CHANGED

WHAT DID NOT CHANGE

Starters: When the axis started at zero, the bars _____. The numbers _____, and _____ was still _____.

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Before you hand it up

Tick each one. If you cannot tick it, go back.

- [ ] Every variable in B1 has a type written beside it.
- [ ] My B2 limitation says what difference it could make, not just what went wrong.
- [ ] Both my displays have a title and labelled axes.
- [ ] My B4 group counts add to the number of measurements I have.
- [ ] My B5 answer names one feature, not two separate descriptions.
- [ ] For every fault in B6 I have said what it makes a reader believe.
- [ ] My B7 answer says what did NOT change as well as what did.
- [ ] My name is on this booklet and on my Part A.

Adjustments applied (teacher use)	

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